

Price Signals and Enrolment Decisions: Evidence from Australia's Job-Ready Graduates Scheme

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Introduction

In the lead-up to a set of education reforms introduced in 2020, there was a perceived mismatch between popular fields of study and areas of job shortage in the Australian economy. The federal government's response was the Job-Ready Graduates Scheme (JRGS), implemented in 2021, which restructured domestic undergraduate tuition fees to incentivise high-priority disciplines (education, health, agriculture, engineering, and IT) through increased commonwealth funding and reduced student contributions, while discouraging low-priority disciplines such as management & commerce, law, arts, and social sciences through the inverse. Before the JRGS, maximum student contributions across disciplines ranged from roughly AUD 6,500 to AUD 11,000 per annum; following the reform, priority disciplines fell to approximately AUD 4,000–9,500 while non-priority disciplines rose to AUD 11,500–16,500, producing a spread of up to AUD 12,500 within a single year of study. Literature on tuition price elasticity suggests demand is inelastic at the market level but elastic when comparing variations across competing institutions (Carter & Curry, 2011), with other factors including location, curriculum, and interpersonal relationships also shaping enrolment decisions (Langelett et al., 2015). Existing evaluations of the JRGS have likened it to 'a misguided arrow missing its target', arguing it is 'unlikely to have a significant effect on student choices' and that disciplinary selection is more heavily driven by socio-economic background, career guidance, and occupational expectations not addressed by the scheme (Daly & Lewis, 2020). However, these evaluations do not examine whether the policy may have succeeded selectively across specific disciplines, a gap this paper addresses by providing the first discipline-level evaluation of the JRGS using a cross-national difference-in-differences design.

This paper tests whether the JRGS produced enrolment shifts consistent with its intended pricing incentives, and whether any observed shifts align with the policy design. The approach uses a difference-in-differences estimator with UK and New Zealand (NZ) university enrolments as treatment-absent control groups. Both represent appropriate comparators: they share similar higher education structures, income-contingent loan schemes, and broadly comparable enrolment patterns, yet neither underwent a discipline-specific fee reform during the study period. The UK's domestic fee cap (£9,250 per annum) has been frozen uniformly across all subjects since 2017, meaning UK enrolment variation by discipline reflects demand and supply factors rather than price incentives. NZ's Fees Free policy (effective 2018) subsidised the first year of study universally across all disciplines, placing it fully in the pre-treatment window and rendering it orthogonal to discipline-level identification. University enrolment panel data spanning 2016–2024 across eleven disciplines for all three countries is used, alongside panel data on Australian university funding clusters and job shortage indicators. The estimating equation applies two-way fixed effects (country and year) with HC3 standard errors, justified by the small number of country clusters ($G=3$).

The findings suggest the JRGS produced a detectable enrolment response in two disciplines: education enrolments increased by approximately 18.6% relative to control countries after 2021 ($p=0.035$, significant at 5%), while management & commerce fell by approximately 21.6% ($p=0.055$, significant at 10% only). The remaining nine disciplines showed no statistically significant change. While the two disciplines with the largest fee changes moved in the direction the JRGS intended, consistent with students responding to price signals, these results alone cannot establish whether this reflects genuine demand elasticity or concurrent confounding factors. The null effects across the remaining disciplines suggest the scheme had limited broader impact in realigning enrolment with labour market priorities.

Data

Data was collected across three countries: Australia (treatment), the United Kingdom (control), and New Zealand (control). Additional enrolment data was collected for Canada as a potential extra control, however it was ultimately excluded from the primary analysis as it introduced excess noise into the panel and was instead retained as an auxiliary robustness check. Panel data was utilised for all three countries.

Three categories of data were collected. First, university enrolments for all three countries spanning 2016–2024 across all disciplines and fields of study, which served as the centrepiece of the difference-in-differences analysis. Second, Australian job market shortage data across a range of industries, which was excluded from the primary econometric analysis and used instead to provide contextual support for the classification of disciplines as high-priority or low-priority under the JRGS. Third, tuition fees and commonwealth funding clusters for Australian tertiary disciplines across 2016–2024, which were matched to their respective disciplines to quantify how student contributions and government co-payments changed as a result of the reform.

Data Cleaning and Sample Restrictions

Disciplinary enrolments across the three countries were assigned common category keys to ensure each Australian treatment discipline had a directly comparable control counterpart in both the UK and New Zealand. The UK HESA data reported enrolments by academic year (e.g. 2016/2017), creating a six-month misalignment with the Australian and New Zealand calendar-year data; this was resolved by mapping each academic year to its start year (e.g. 2016/2017 became 2016), producing a consistent annual time index across all three countries. Australian funding clusters were also standardised across all time periods, as different versions of the raw data classified certain disciplines across different clusters; these were corrected to ensure that funding figures accurately reflected each discipline's actual allocation throughout the study window. The same category key system was then used to link funding clusters to their corresponding disciplines.

Certain disciplines were not consistently categorised in the raw data for the years 2016–2018, particularly for UK subject classifications which changed in 2019/2020. Observations for these disciplines prior to 2019 were dropped to preserve comparability, limiting five disciplines (natural and physical sciences, environment and related, society and culture, others, and architecture and building) to a 2019–2024 panel. The final observation count per discipline is $N=27$ ($3 \text{ countries} \times 9 \text{ years}$) for disciplines with the full 2016–2024 panel, and $N=18$ ($3 \text{ countries} \times 6 \text{ years}$) for disciplines restricted to 2019–2024. A log transformation was applied to the enrolment outcome variable to express results as percentage changes and to accommodate the substantial differences in nominal enrolment scale across the three countries. Together, these adjustments produce a balanced and comparable panel in which the primary source of structural variation is the JRGS shock rather than differences in national counting conventions.

Treatment Intensity by Discipline

Table D1. Maximum Student Contribution by Discipline — Pre- and Post-JRGS (AUD per EFTSL)

Discipline	JRGS	Pre-JRG (avg 2019–20)	2021	2024	Change (\$)	Change (%)
Management & Commerce	Non-priority	\$11,056	\$14,500	\$16,323	+3,444	+31.1%
Creative Arts	Non-priority	\$6,625	\$9,522	\$10,718	+2,897	+43.7%
Society & Culture	Non-priority	\$7,364	\$11,636	\$13,099	+4,273	+58.0%
Health	Priority	\$9,462	\$8,451	\$9,512	-1,011	-10.7%
Information Technology	Priority	\$9,443	\$7,950	\$8,948	-1,493	-15.8%
Engineering & Related Tech	Priority	\$9,443	\$7,950	\$8,948	-1,493	-15.8%
Architecture & Building	Priority	\$9,443	\$7,950	\$8,948	-1,493	-15.8%
Natural & Physical Science	Priority	\$9,487	\$7,518	\$8,461	-1,969	-20.8%
Education	Priority	\$6,625	\$3,950	\$4,445	-2,675	-40.4%
Environment & Related	Priority	\$9,443	\$5,213	\$5,867	-4,230	-44.8%

Source: Australian Government Department of Education, Annual Funding 2019–2026.

The JRGS produced a wide dispersion in fee changes across disciplines that had previously been relatively compressed. Before the reform, maximum student contributions ranged from approximately AUD 6,500 to AUD 11,000 per equivalent full-time student load. By 2021, that range had widened to roughly AUD 4,000–15,000, with priority disciplines receiving cuts of between 20% and 46% and non-priority disciplines absorbing increases of 28–36%. Education received the largest proportional cut among priority fields, while management and commerce faced the steepest increase among non-priority fields; these are also the two disciplines that produce the only statistically significant results in the analysis. The variation in treatment intensity across disciplines provides a natural dose-response test: if fee signals drive enrolment decisions, the largest effects should be concentrated in the disciplines with the largest fee changes, which is broadly consistent with the findings.

Summary Statistics

Table D2. Summary Statistics — Enrolment Panel (AUS, UK, NZ; all disciplines)

Discipline	Panel	N	Mean enrolments	SD enrolments	Mean log(enrol)	SD log(enrol)	Pre-2021 mean log	Post-2021 mean log
Natural & Physical Science	Short (2019–2024)	30	79,874	39,617	11.162	0.519	11.171	11.157
Information Technology	Short (2019–2024)	18	106,953	68,546	11.225	1.003	11.133	11.270
Engineering & Related Tech	Full (2016–2024)	27	99,609	71,466	10.855	1.474	10.868	10.838
Architecture & Building	Full (2016–2024)	27	34,915	24,332	9.947	1.251	9.887	10.021
Environment & Related	Short (2019–2024)	30	17,850	9,169	9.639	0.593	9.621	9.647
Health	Full (2016–2024)	45	143,745	132,375	11.261	1.238	11.227	11.303
Education	Full (2016–2024)	27	93,750	58,845	11.026	1.123	11.034	11.016
Management & Commerce	Full (2016–2024)	27	295,223	196,631	12.171	1.110	12.139	12.210
Society & Culture	Short (2019–2024)	48	147,098	99,254	11.692	0.640	11.695	11.691
Creative Arts	Full (2016–2024)	27	101,065	66,649	11.223	0.868	11.235	11.208
Others	Short (2019–2024)	24	18,220	19,351	8.426	2.087	8.522	8.378

Units: enrolments = headcount; log(enrolments) = natural log of annual headcount. Total observations: 330 | Countries: AUS (treated), UK (control), NZ (control). Full panel (N=27 each): Engineering & Related Tech, Architecture & Building, Health, Education, Management & Commerce, Creative Arts. Short panel (N=18 each): Natural & Physical Science, Information Technology, Environment & Related, Society & Culture, Others.

Sample Construction and Exclusions

Table D0. Sample Construction

Step	Dropped	Remaining
Raw data collected	—	538
Restricted to 2016–2024 study window	79	459
Restricted to 11 matched category keys	45	414
Dropped pre-2019 for 5 UK-reclassified disciplines	84	330
Dropped rows with missing enrolment values	0	330

Final analytic sample: 330 observations. 6 disciplines × 3 countries × 9 years = 162 (full panel, N=27 each); 5 disciplines × 3 countries × 6 years = 90 (short panel, N=18 each).

Empirical Strategy

The hypothesis is that the JRGS fee changes shifted enrolments toward high-priority disciplines and away from those facing fee increases. The data structure is designed to isolate this effect: Australia is the only country in the panel to have altered its higher-education fee structure, while the UK and NZ held their funding arrangements effectively constant over the 2021–2024 period. The panel structure ensures that within-country, within-discipline variation over time drives the estimation rather than permanent scale differences between Australia and the control countries. The JRGS treatment also varies in intensity across disciplines, and disciplines that experienced larger fee movements are expected to exhibit larger enrolment responses if a treatment effect exists. Year fixed effects absorb any global shocks common to all three countries in a given year, most notably the disruptions associated with COVID-19, leaving only the Australia-specific post-2021 divergence in enrolments to be attributed to the JRGS.

Methodology

The estimating equation is a two-way fixed effects difference-in-differences model, estimated separately for each discipline:

$$\log E_{ct} = \alpha + \beta_1 \text{treated}_c + \beta_2 \text{NZ}_c + \beta_3 (\text{treated}_c \times \text{post}_t) + \sum_{t=2017}^{2024} \gamma_t \mathbf{1}[t] + \varepsilon_{ct}$$

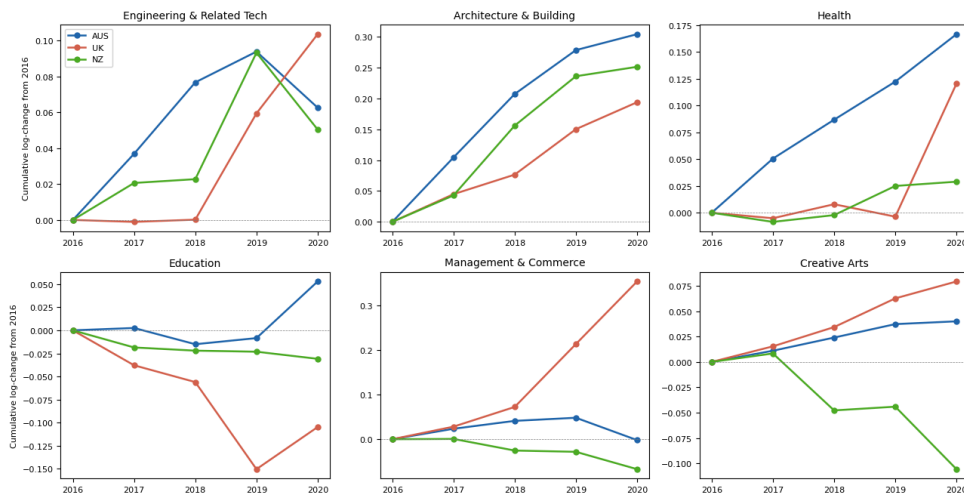
where $c \in \{\text{AUS, UK, NZ}\}$ and t indexes calendar years. treated_c and NZ_c are country indicators that absorb permanent level differences across the three countries, with the UK as the reference unit. $\text{post}_t = \mathbf{1}[t \geq 2021]$ marks the JRGS implementation period. Year fixed effects γ_t absorb shocks common to all three countries in a given year, including the global disruption from COVID-19. The outcome variable is log-transformed for two reasons. Enrolment counts vary substantially in absolute scale across disciplines — health and nursing enrol roughly ten times as many students as architecture or environment — so raw-count differences would conflate scale with response. Logs render the coefficient interpretable as a proportional change and stabilise residual variance across disciplines of different sizes, satisfying the assumptions underlying HC3 inference. The log-linear specification is also standard in the enrolment demand literature.

The coefficient of interest, $\hat{\beta}_3$, measures the log-point difference-in-differences: the post-2021 deviation of Australian log-enrolments from the counterfactual trajectory implied by the pooled UK–NZ trend, net of any pre-existing level difference between Australia and the controls captured by β_1 and β_2 . The exact implied percentage change in enrolments is $(e^{\hat{\beta}_3} - 1) \times 100$; for the coefficient magnitudes observed here ($|\hat{\beta}_3| < 0.3$), this is well approximated by $100 \times \hat{\beta}_3$ percentage points. A positive $\hat{\beta}_3$ indicates Australian enrolments in that discipline grew faster than the UK–NZ trend post-2021; a negative estimate indicates relative decline.

Standard errors are HC3 throughout. With $G=3$ country clusters, cluster-robust inference is not feasible: the Moulton factor for this panel implies non-trivial within-cluster error correlation, and the minimum cluster count for asymptotically reliable cluster-robust SEs is approximately 42. HC3 provides a conservative alternative. The primary specification leaves 15 residual degrees of freedom for full-panel disciplines ($N=27$) and 9 for short-panel disciplines ($N=18$).

The identifying assumption is that, absent the JRGS, Australian enrolments in each discipline would have followed a trend parallel to the pooled UK and NZ counterfactual from 2021 onward. This is evaluated below using pre-treatment trends for the six full-panel disciplines, where five years of pre-treatment data are available.

Figure 1. Pre-treatment log-enrolment trends — full-panel disciplines, indexed to 2016 (AUS, UK, NZ)



Pre-treatment trends are broadly parallel for most full-panel disciplines, though with meaningful variation. Management and commerce and health track closely across all three countries through 2016–2020, with deviations from a common slope remaining small throughout the pre-treatment window, and these two disciplines provide the strongest support for the parallel trends assumption. Creative arts also shows close co-movement, though the small absolute enrolment numbers in NZ produce more volatility in the indexed series.

Education shows a mild but consistent upward divergence of AUS from the controls from 2017 onward, with Australian enrolments growing at a faster rate than both the UK and NZ in the pre-treatment period. While not large in magnitude, this pre-existing divergence is in the same direction as the positive post-treatment DiD estimate, which is consistent with the sign reversal observed when country-specific linear trends are added to the main specification. It suggests the education result may partly reflect a continuation of a pre-existing Australian trend rather than a discrete post-2021 response to the fee change. Engineering shows a similar pattern of AUS outpacing the controls from 2018, introducing analogous concerns for that discipline's estimate. Architecture is the most clearly parallel through 2019 but shows AUS enrolments plateauing while UK and NZ continue growing into 2020, which complicates the post-treatment counterfactual. The pre-treatment window is limited to 2019–2020 for the five short-panel disciplines, which is insufficient to meaningfully assess the parallel trends assumption for those fields.

Results

Main Results

Table 1 reports the TWFE DiD coefficient (β_3 on the interaction term $\text{treated} \times \text{post}$) for all eleven disciplines, estimated with HC3 standard errors.

Table 1. Effect of the Job-Ready Graduates Scheme on University Enrolments

Dependent variable: $\log(\text{annual enrolments})$, Australia vs UK + New Zealand panel. Estimating equation: $\log(\text{enrolments}) \sim \text{treated} + \text{nz_dummy} + \text{did} + C(\text{year})$, HC3 SEs.

Panel A — Priority disciplines (JRG fee reduction)

	Education	Eng. & RT	Health	IT	Nat. Sci.	Environment
DiD (β_3)	+0.171*	+0.019	-0.011	+0.141	+0.009	-0.068
	(0.081)	(0.061)	(0.039)	(0.160)	(0.022)	(0.083)
Approx. %	+18.6%	+1.9%	-1.1%	+15.1%	+0.9%	-6.5%
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	27	27	27	27	18	18
Adj. R ²	0.997	0.998	0.999	0.968	1.000	0.999

Panel B — Discouraged disciplines (JRG fee increase)

	Mgmt & C	Soc. & Cult.
DiD (β_3)	-0.243†	+0.008
	(0.127)	(0.037)
Approx. %	-21.6%	+0.8%
Year fixed effects	Yes	Yes
Country fixed effects	Yes	Yes
<i>N</i>	27	18
Adj. R ²	0.979	0.999

Panel C — Non-priority disciplines

	Arch. & Build.	Creative Arts	Others
DiD (β_3)	-0.054	+0.026	-0.268
	(0.033)	(0.035)	(0.241)
Approx. %	-5.2%	+2.6%	-23.5%
Year fixed effects	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes
<i>N</i>	27	27	18
Adj. R ²	0.999	0.998	0.993

Notes: HC3 heteroskedasticity-robust standard errors in parentheses. *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$ † $p < 0.10$. Country fixed effects absorbed via *treated* (AUS=1) and *nz_dummy* (NZ=1); year fixed effects absorb common shocks including COVID-19. *N*=27 for 2016–2024 panel; *N*=18 for 2019–2024 panel (five disciplines). Approx. % = $(\exp(\beta) - 1) \times 100$. Data: AUS DESE; UK HESA; NZ TEC.

The JRGS fee reform is associated with a meaningful enrolment response in two disciplines, Education and Management & Commerce, and no detectable effect in the remaining nine.

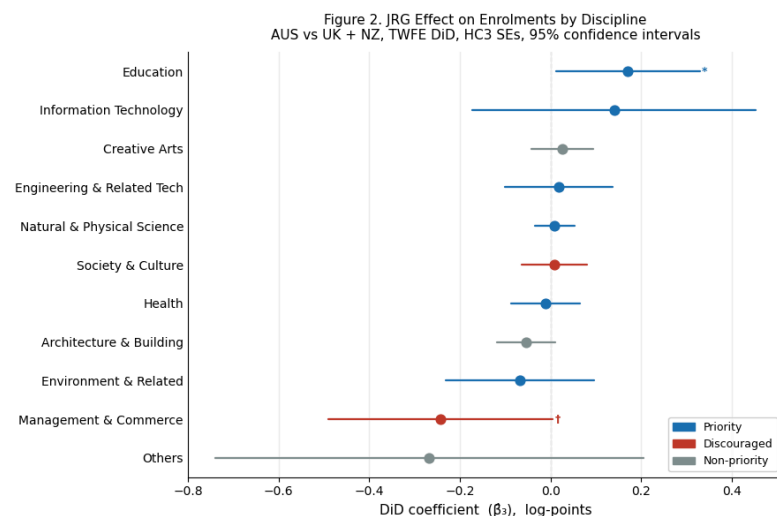
Education (Panel A), where student fees were reduced by approximately 70%, shows a positive and statistically significant DiD coefficient: $\beta_3 = +0.171$ (SE = 0.081, $p = 0.035$), implying approximately 18.6% higher Australian enrolments relative to the UK and New Zealand trend following the 2021 reform. The result survives most robustness checks but should be treated as an upper bound: UK Education enrolments were independently declining post-2021 due to domestic teacher training policy changes, and the coefficient reverses sign when country-specific linear trends are controlled for.

Management & Commerce (Panel B), where fees approximately doubled, shows the opposite: $\beta_3 = -0.243$ (SE = 0.127, $p = 0.055$), implying a 21.6% relative decline in Australian enrolments against trend. This result is only marginally significant and should be treated as suggestive rather than cleanly identified: both wrong-group placebos register significant post-2021 movements in the control countries, and the coefficient collapses to insignificance when two UK pre-treatment observations are dropped.

The remaining nine disciplines produce coefficients that are uniformly small and statistically insignificant (all $p \geq 0.10$). The two disciplines whose fee structures changed most dramatically moved in the expected direction; fields with smaller or no fee changes did not. However, the robustness evidence tempers a strong causal reading: post-2021 global dynamics in international student demand and pre-existing country-specific trends make clean attribution difficult, particularly for Management & Commerce.

Coefficient Plot

Figure 2 plots DiD coefficient estimates (β_3) and 95% confidence intervals for all eleven disciplines, sorted by effect size. Priority disciplines receiving fee reductions under the JRGS are shown in blue, discouraged disciplines facing fee increases in red, and non-priority disciplines in grey. The horizontal axis measures log-point deviations of Australian enrolments from the pooled UK-NZ counterfactual trend post-2021; the dashed line at zero indicates no change relative to trend.



Education stands out clearly among Priority fields. The five other Priority disciplines (Health, Engineering, IT, Environment, Natural Sciences) cluster near zero, while Education alone produces an estimate with its interval lying largely above zero. The Education fee cut, at approximately 70% the most dramatic among Priority fields, produced the clearest enrolment signal.

Among Discouraged disciplines, Management & Commerce is the only field with a notable negative estimate. Society & Culture shows a coefficient close to zero, suggesting the JRGS fee increase had a more pronounced effect on M&C than on its counterpart Discouraged field, though the M&C result carries the identification caveats described above.

The null results across nine disciplines are credible rather than simply imprecise: confidence intervals are narrow and centred near zero. The Education and M&C estimates are of broadly similar absolute size (approximately 0.17 and 0.24 log-points respectively), consistent with roughly symmetric responsiveness to fee changes in opposite directions.

Robustness

Table 2 summarises the key specification checks for the two significant estimates. Each row reports the DiD coefficient under an alternative sample restriction, control configuration, or placebo assignment. The education and management and commerce results respond very differently under scrutiny.

Table 2. Robustness of Main Estimates

Specification check	Education	Management & Commerce
Baseline (HC3)	+0.171*	-0.243†
Exclude 2020	+0.188**	-0.271*
Exclude 2020 and 2021	+0.218**	—
Country-specific trends	-0.043	—
UK as sole control	Positive*	-0.501***
NZ as sole control	Positive†	+0.015
Remove high-leverage obs.	+0.133†	-0.161
UK wrong-group placebo	-0.172*	+0.508***
NZ wrong-group placebo	—	-0.265*
Permutation p-value	0.667	1.000

Dependent variable: log(annual enrolments). Coefficient of interest: DiD (treated × post). *** p<0.001 ** p<0.01 * p<0.05 † p<0.10. Baseline uses HC3 standard errors with G=3 country clusters. “Positive” indicates sign preserved without reporting the exact coefficient (not reported in original checks). Permutation p-value is the share of treatment assignments (out of 3) producing a larger absolute coefficient than the true Australian assignment. — indicates the check was not performed for that discipline.

The education estimate is stable across sample restrictions. Excluding the COVID baseline year or both COVID years strengthens the result, indicating it does not depend on the disruption period, and the sign is preserved across UK-only and NZ-only jackknife configurations. Two checks raise substantive concerns. The UK wrong-group placebo is statistically significant ($\beta = -0.172$, $p = 0.035$), reflecting an independent post-2021 decline in UK education enrolments due to domestic teacher training policy changes, meaning part of the estimated AUS–UK divergence reflects movement in the control group rather than a genuine Australian increase. The coefficient also reverses sign when country-specific linear trends are incorporated ($\beta = -0.043$), consistent with the pre-treatment data showing a mild upward divergence of Australian education enrolments from 2017 onward. Removing three high-leverage observations weakens but preserves the sign ($\beta = +0.133$, $p = 0.064$). The education estimate is best treated as an upper bound.

The management and commerce estimate is less stable. The coefficient strengthens under COVID exclusion and is notably larger under UK-only identification ($\beta = -0.501$, $p < 0.001$), but collapses to near zero in the NZ-only specification ($\beta = +0.015$), revealing that the pooled result is driven almost entirely by the AUS–UK comparison. Both wrong-group placebos are statistically significant, and the permutation test places the true Australian assignment last among all three possible allocations (permutation $p = 1.00$). Dropping two high-leverage UK pre-treatment observations reduces the estimate to -0.161 ($p = 0.181$). The M&C result is directionally consistent with the policy but cannot be given a credible causal interpretation under this design.

Discussion, Limitations, and Conclusion

The results neither fully validate nor fully dismiss the prior literature on JRGS effectiveness. Nine of eleven disciplines produce null results, consistent with the Daly and Lewis (2020) characterisation of the scheme and the broader finding that tuition price demand is inelastic at the market level (Havranek et al., 2018; Langelett et al., 2015). The two exceptions, education and management and commerce, are the fields where the JRGS moved student contributions furthest outside the historical range: below the prior fee floor in the case of education and above the prior ceiling in the case of management and commerce. Disciplines whose contributions were repositioned within the previously familiar bandwidth produced no detectable response. This is consistent with the inter-institutional price competition mechanism proposed by Carter and Curry (2011), applied across disciplines within the same institution rather than between competing universities, and suggests a salience threshold at which fee changes become legible to prospective students.

The null results in health, engineering, and information technology are particularly notable given that these fields combine the most substantial fee reductions with the most acute documented labour market shortages. Entry into these vocational professions is governed primarily by career commitment, prior subject investment, and intrinsic motivation that deferred fee changes operating through HECS-HELP are unlikely to override (Janke, 2020). The fee reductions in these fields most likely functioned as transfers to students who would have enrolled regardless rather than as supply-expanding incentives.

Evaluated against the JRGS's stated objectives, the results suggest the policy achieved its enrolment goals in two disciplines but not the ones that matter most for the labour market problem it was designed to address. Education is a genuine shortage field and the observed increase, if attributable to the policy, is directionally consistent with the scheme's intent. Management and commerce is a field with persistently low shortage rates, and a decline in enrolments from a surplus-supply discipline is directionally appropriate. The null results across the remaining nine disciplines, however, include fields with sustained shortage rates above 70% of relevant occupations: the policy succeeded in redirecting students away from a low-shortage field while failing precisely where the labour market need was greatest.

As detailed in Table 2, both headline estimates face substantive challenges. The education coefficient is best treated as an upper bound, given a significant UK wrong-group placebo and a sign reversal under country-specific linear trends. The management and commerce estimate collapses to near zero under NZ-only identification and is ranked last in permutation inference, indicating it cannot be credibly separated from broader post-2021 dynamics affecting anglophone higher education systems.

Limitations

The three-country cluster structure ($G=3$) precludes reliable cluster-robust standard errors; HC3 inference and permutation testing serve as partial remedies but cannot achieve conventional significance thresholds at this cluster size. Annual institution-level data prevents identification of compositional changes within disciplines, and the inability to separate domestic from international enrolments in the Australian source data may attenuate estimated treatment effects, since international students were unaffected by the JRGS funding changes. The post-treatment window extends only to 2024, with the 2021 cohort not yet fully graduated, and any demand-side enrolment shifts may not yet have translated into the graduate supply the scheme intended to produce.

Conclusion

The JRGS produced enrolment shifts consistent with its pricing incentives in education and management and commerce, concentrated at the extremes of its fee structure where changes were large enough to breach the historically familiar range of student contributions. The remaining nine disciplines, including those with the most acute labour market shortages and the most substantial fee reductions, showed no significant enrolment response, consistent with enrolment decisions in vocational fields being governed primarily by non-financial motivations. Both significant estimates are materially constrained by robustness evidence and cannot be read as clean causal estimates of the JRGS effect. Future evaluation would benefit from individual-level data disaggregating domestic from international enrolments, longer post-treatment panels as the 2021 cohort matures toward graduation, and analysis of completion and employment outcomes to determine whether observed demand shifts translated into the graduate supply the scheme intended to produce.

References

- Carter, R. E., & Curry, D. J. (2011). Using student-choice behaviour to estimate tuition elasticity in higher education. *Journal of Marketing Management*, 27(11-12), 1186-1207. <https://doi.org/10.1080/0267257X.2011.609653>
- Daly, A., & Lewis, P. (2020). The proposed Job-ready Graduate Package: A misguided arrow missing its target. *Australian Journal of Labour Economics*, 23(2), 231-251.
- Havranek, T., Irsova, Z., & Zeynalova, O. (2018). Tuition fees and university enrolment: A meta-regression analysis. *Oxford Bulletin of Economics and Statistics*, 80(6), 1145-1184. <https://doi.org/10.1111/obes.12240>
- Janke, S. (2020). Prospective effects of motivation for enrollment on well-being and motivation at university. *Studies in Higher Education*, 45(12), 2413-2425. <https://doi.org/10.1080/03075079.2019.1612353>
- Langelett, G., Chang, K.-L., Ola' Akinfenwa, S., Jorgensen, N., & Bhattarai, K. (2015). Elasticity of demand for tuition fees at an institution of higher education. *Journal of Higher Education Policy and Management*, 37(1), 111-119. <https://doi.org/10.1080/1360080X.2014.991536>

Data Sources

- Australian Government Department of Education. (2016-2024). *University enrolments by field of education* [PowerBI dashboard]. Retrieved from <https://app.powerbi.com/view?r=eyJrJoiN2Y1ZTM1Y2YtYzYxNC00MWI0LWFhNTktYTI3NWU2OWI3NGFklwidCI6ImRkMGNmZDE1LTQ1NTgtNGlxMi04YmFkLWVhMjY>
- Australian Government Department of Education. (2019-2026). *Funding clusters and indexed rates: Commonwealth and maximum student contributions by discipline*. Retrieved from <https://www.education.gov.au/higher-education-loan-program/approved-hep-information/funding-clusters-and-indexed-rates#toc-indexed-rates>
- Department of Employment and Workplace Relations. (2021-2025). *Occupation Shortage List*. Jobs and Skills Australia, Canberra. Retrieved from <https://www.jobsandskills.gov.au/data/occupation-shortage/occupation-shortage-list>
- Higher Education Statistics Agency (HESA). (2016-2024). *Higher education student statistics: Subjects*. Retrieved from <https://www.hesa.ac.uk/news/27-01-2026/sb273-higher-education-student-statistics/subjects>
- Ministry of Education New Zealand. (2016-2025). *Provider-based enrolments: Field of study at the course level*. Education Counts. Retrieved from <https://www.educationcounts.govt.nz/statistics/tertiary-participation>
- Statistics Canada. (2016-2024). *Postsecondary enrolments, by field of study, registration status, program type, credential type and gender* [Table 37-10-0011-01]. Retrieved from <https://www150.statcan.gc.ca/t1/tbl1/en/tbl/dtbl/37100011>